Project 1: Data Science & Engineering Python Project

Project Brief Email

From: engineering-leads@nexus-analytics.com
To: Project Engineering Consultant
Subject: Request for Services: Production Line Predictive Maintenance Data Pipeline

Dear Consultant,

Our manufacturing client is experiencing unplanned downtime on their primary conveyor assembly systems. They collect hourly telemetry data containing operational metrics (temperature, vibration, operational hours) but lack an automated way to clean this data, flag imminent anomalies, and export clean metrics for downstream dashboarding.

We require you to develop a robust, production-ready Python data pipeline that reads a raw dataset, cleans missing entries, engineers statistical features, isolates out-of-bounds metrics, and outputs a formatted engineering summary.

Please deliver the structured Python pipeline code along with an engineering implementation report.

Best regards,
Nexus Analytics Engineering Team